

Youcheng Li

Co-founder & CTO, Isoplex Intelligence

Ph.D. Student in Artificial Intelligence, Peking University

Advisor: Prof. Liwei Wang

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Profile

Research in **medical AI**, **generative foundation models**, and **scientific agents**, translating research into products and scientific workflows. First-author and co-first-author publications in **Nature Biomedical Engineering**, **Scientific Data**, **KDD**, and **PLOS Computational Biology**. As Co-founder and CTO of Isoplex Intelligence, lead research workspaces, scientific world models, and experimental feedback systems, including technical development, team building, and industry delivery.

Entrepreneurship & Industry Experience

Isoplex Intelligence (壹索智能)

Jun 2025 – Present Beijing

Co-founder & CTO | 北京壹索智能科技有限公司

- **Strategy and teams:** Lead technical strategy and product planning; build and manage agent, frontend, and backend engineering teams, taking products from MVP to continuous iteration and customer delivery.
- **Research workspace (Iso-Sciland):** Lead the architecture and development of the AI Scientist agent system, integrating tool use, scientific data analysis, and research collaboration into researchers' daily workflows.
- **Scientific world models (Iso-SWM):** Lead the technical roadmap, model training, and evaluation for multimodal scientific state representations, intervention simulation, and research decision-making.
- **Physical AI and experimental feedback (Iso-LabOS):** Lead the roadmap, system architecture, and validation prototypes connecting model predictions, experimental execution, and result feedback.
- **Deployment and financing:** Drive deployments at Grade-A tertiary hospitals, universities, and listed pharmaceutical companies; contribute to business development and fundraising, helping secure **tens of millions of RMB in seed financing**.

YIZHUN AI

Jan 2023 – Jun 2025 Beijing

Algorithm Research Intern

- **Diagnostic and generative models:** Researched breast ultrasound screening, diagnosis, and synthetic data; work published at MICCAI 2023 and, as co-first author, in *Nature Biomedical Engineering* (2026).
- **Real-time detection:** Developed lesion detection for breast ultrasound video with sensitivity > 95%; reduced inference latency by **15%** through model compression and hardware acceleration.
- **Clinical applications:** Developed AI diagnosis algorithms for ductal carcinoma in situ (DCIS); the project won **National First Prize** at the 2nd National Digital Health Innovation Application Competition.

Early Research & Engineering Experience

Western University

Sep 2022 – Jul 2023 Remote

Research Assistant | Advisor: Prof. Pingzhao Hu

Supported by Mitacs and the China Scholarship Council. Proposed **ST-CellSeg** for spatial transcriptomics cell segmentation, implemented the algorithm, and conducted systematic benchmarks; published as **first author** in *PLOS Computational Biology*.

Shenzhen Anke High-tech Co., Ltd.

Jul 2020 – Sep 2020

Assistant Embedded Systems Engineer (Intern)

Contributed to medical-device circuit debugging, firmware optimization, and ARM-based embedded system integration testing.

Selected Publications

[†] denotes equal contribution; name in bold. Titles link to the publications.

Yu H[†], **Li Y[†]**, Zhang N[†], Niu Z[†], et al. **Nature Biomedical Engineering**, 2026
A foundation generative model for breast ultrasound image analysis
BUSGen: A generative foundation model pretrained on over 3.5 million breast ultrasound images for screening, diagnosis, and prognosis.

Dai D[†], Liu B[†], **Li Y[†]**, Yu H[†], et al. **KDD 2026, AI4Sciences**, 10785–10794
MammoExpert: Benchmarking Chain-of-Thought Reasoning in Mammography Diagnosis
MammoExpert: A diagnostic reasoning benchmark with 2,379 mammograms spanning 67 pathology types.

Yu H[†], **Li Y[†]**, Niu Z[†], et al. **Scientific Data**, 2026, 13: 370
A Chain-of-thought Reasoning Breast Ultrasound Dataset Covering All Histopathology Categories
BUS-CoT: An ultrasound dataset covering 99 WHO pathology categories with structured diagnostic reasoning annotations.

Li Y, Lac L, Liu Q, Hu P. **PLOS Computational Biology**, 2024, 20(6): e1012254
ST-CellSeg: Cell segmentation for imaging-based spatial transcriptomics using multi-scale manifold learning
ST-CellSeg: Multi-scale manifold learning for cell segmentation in imaging-based spatial transcriptomics.

Other publications

Yu H, **Li Y**, Wu Q, et al. Mining Negative Temporal Contexts for False Positive Suppression in Real-Time Ultrasound Lesion Detection. **MICCAI 2023**, 3–13.

Yan H, Dai C, Xu X, ..., **Li Y**, Bao L. Using artificial intelligence system for assisting the classification of breast ultrasound glandular tissue components in dense breast tissue. **Scientific Reports**, 2025, 15: 11754.

Preprint

Yu H, **Li Y**, Zhang N, et al. Knowledge-driven AI-generated data for accurate and interpretable breast ultrasound diagnoses. arXiv:2407.16634, 2024.

Education

Peking University	Ph.D. in Artificial Intelligence (in progress)	Sep 2023 – Present
	School of Intelligence Science and Technology; advisor: Prof. Liwei Wang	Expected Jul 2028
Xi'an Jiaotong University	B.Eng. in Artificial Intelligence	Sep 2019 – Jul 2023
	GPA: 4.0/4.3	

Scholarships & Honors

BYD Scholarship; Merit Student	Peking University	2025–2026
Huawei Scholarship	Peking University	2024–2025
National Scholarship; Merit Student	Peking University	2023–2024
Outstanding Graduate	Xi'an Jiaotong University	2022–2023
China Scholarship Council Scholarship; Mitacs Scholarship		2021–2022
MEGVII Scholarship	Xi'an Jiaotong University	2020–2021
National Scholarship; Merit Student	Xi'an Jiaotong University	2019–2020
National First Prize	2nd National Digital Health Innovation Application Competition	2024

Years denote academic years, except the competition award. Other scholarship: Zheng Guobin Scholarship.

Core Technical Skills

Models and algorithms: PyTorch, diffusion models / GANs, medical image analysis, model training and evaluation, model compression and quantization.

Agents and systems: LangChain / LangGraph, multi-agent collaboration, RAG, toolchain integration; Python, C++, JavaScript, SQL.

Engineering and delivery: TensorRT, ONNX, Docker, Linux, Git; hardware-accelerated inference, ARM / RTOS; cross-functional team management and product delivery.